Chapter 5. Approximating Sampling Distributions

Central limit theorem, convergence in distribution and probability, limit computations, simulation

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Aim

Exact sampling distributions (Chapter 4) are rarely available. This chapter develops the large-sample tools that replace them: convergence in distribution and in probability, the central limit theorem, Slutsky's theorem and the delta method. It ends with simulation.

5.1 The central limit theorem

Theorem 5.1 (Central limit theorem)

Let $X_1, X_2, \dots$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Then for every x,

$$P\Big(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \le x\Big) \to \Phi(x).$$

Proof assuming an mgf. Let $Y_i = (X_i - \mu)/\sigma$ with mgf M_Y near 0; then $M_Y(t) = 1 + \frac{t^2}{2} + o(t^2)$. The mgf of $Z_n = \frac{1}{\sqrt{n}} \sum Y_i$ is $M_Y(t/\sqrt{n})^n = \big(1 + \frac{t^2}{2n} + o(\frac{1}{n})\big)^n \to e^{t^2/2}$, the $N(0,1)$ mgf. Conclude by the continuity theorem (Theorem 5.2). Without an mgf the same argument works with characteristic functions. ■

- **Normal approximation to the binomial** (de Moivre–Laplace): $X \sim B(n,p)$ is a sum of Bernoullis, so $(X - np)/\sqrt{np(1-p)} \xrightarrow{d} N(0,1)$. For integer x use the continuity correction $P(X \le x) \approx \Phi\big((x + \frac{1}{2} - np)/\sqrt{np(1-p)}\big)$. Similarly for Poisson($\lambda$) as $\lambda \to \infty$, and for $\chi^2(r)$ as $r \to \infty$: $(\chi^2(r) - r)/\sqrt{2r} \xrightarrow{d} N(0,1)$.
- **Multivariate CLT.** For i.i.d. random vectors with mean μ and covariance Σ, $\sqrt{n}(\bar{\mathbf{X}}_n - \mu) \xrightarrow{d} N_k(\mathbf{0}, \Sigma)$. For multinomial counts, $(\mathbf{X}_n - n\mathbf{p})/\sqrt{n} \xrightarrow{d} N_k(\mathbf{0}, \mathrm{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}')$, a singular normal since the counts sum to n.

Example 5.1

For $X \sim B(100, 0.5)$: $P(X \le 55) \approx \Phi\big(\frac{55.5-50}{5}\big) = \Phi(1.1) = 0.864$ (exact: 0.864); without the correction, $\Phi(1) = 0.841$.

5.2 Limiting distributions and convergence in probability

Definition (Convergence in distribution)

$X_n \xrightarrow{d} Z$ if $P(X_n \le x) \to P(Z \le x)$ at every continuity point x of the cdf of Z. The law of Z is the **limiting** (asymptotic) distribution.

Continuity points matter. If X_n is uniform on $(0, 1 + \frac{1}{n})$ with probability $\frac{1}{2}$ and equals $1 + \frac{1}{n}$ otherwise, the cdfs converge to $\frac{x}{2}$ on $[0,1)$ and to 1 for $x > 1$, but at $x = 1$ the limit is $\frac{1}{2}$. The natural limit Z (uniform on $(0,1)$ with probability $\frac{1}{2}$, the value 1 otherwise) has cdf 1 at 1, and $X_n \xrightarrow{d} Z$ even though convergence fails at that one point.

Theorem 5.2 (Continuity theorem)

If $M_{X_n}(t) \to M_Z(t)$ for all t in a neighborhood of 0, then $X_n \xrightarrow{d} Z$. (Proof omitted; it is the mgf analogue of Lévy's theorem.)

Definition (Convergence in probability)

$X_n \xrightarrow{p} c$ (written $\operatorname{plim} X_n = c$) if $P(|X_n - c| \geq \epsilon) \to 0$ for every $\epsilon > 0$. For vectors, use the norm $\|\mathbf{X}_n - \mathbf{c}\|$.

Theorem 5.3

(a) $X_n \xrightarrow{p} c$ iff $X_n \xrightarrow{d} c$ (the constant). (b) If $E(X_n - c)^2 \to 0$ then $X_n \xrightarrow{p} c$. (c) A vector converges in probability iff each coordinate does.

Proof. (a) The cdf of the constant c is continuous except at c. If $X_n \xrightarrow{p} c$, then for $x < c$, $P(X_n \leq x) \leq P(|X_n - c| \geq c - x) \to 0$, and for $x > c$, $P(X_n \leq x) \geq 1 - P(|X_n - c| \geq x - c) \to 1$. Conversely $P(|X_n - c| \geq \epsilon) \leq P(X_n \leq c - \epsilon) + 1 - P(X_n \leq c + \epsilon/2) \to 0$. (b) Markov's inequality with $r = 2$. (c) $|X_{ni} - c_i| \leq \|\mathbf{X}_n - \mathbf{c}\| \leq \sum_i |X_{ni} - c_i|$. ■

Theorem 5.4 (Weak law of large numbers)

If X_i are i.i.d. with $E|X_1| < \infty$, then $\bar{X}_n \xrightarrow{p} EX_1$.

Proof under finite variance. $E(\bar{X}_n - \mu)^2 = \sigma^2/n \to 0$; apply Theorem 5.3(b). (The finite-mean case uses truncation or characteristic functions.) ■

So sample moments are consistent: $\frac{1}{n}\sum X_i^r \xrightarrow{p} EX_1^r$ whenever $E|X_1|^r < \infty$.

Theorem 5.5 (Continuous mapping)

If $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$ and g is continuous at $\mathbf{c}$, then $g(\mathbf{X}_n) \xrightarrow{p} g(\mathbf{c})$. In particular plim commutes with $+$, $-$, $\times$, and with $\div$ when the denominator's limit is nonzero.

Proof. Given $\epsilon > 0$ pick δ with $\|\mathbf{x} - \mathbf{c}\| < \delta \Rightarrow |g(\mathbf{x}) - g(\mathbf{c})| < \epsilon$. Then $P(|g(\mathbf{X}_n) - g(\mathbf{c})| \geq \epsilon) \leq P(\|\mathbf{X}_n - \mathbf{c}\| \geq \delta) \to 0$. For the arithmetic rules, $(X_n, Y_n) \xrightarrow{p} (a, b)$ by Theorem 5.3(c), and $x + y$, xy, x/y are continuous there. ■

Hence $S_n^2 = \frac{n}{n-1}(\frac{1}{n}\sum X_i^2 - \bar{X}_n^2) \xrightarrow{p} \sigma^2$ and $S_n \xrightarrow{p} \sigma$.

5.3 Computing limiting distributions

Theorem 5.6 (Slutsky)

If $X_n \xrightarrow{d} Z$ and $Y_n \xrightarrow{p} c$, then (a) $X_n + Y_n \xrightarrow{d} Z + c$; (b) $Y_n X_n \xrightarrow{d} cZ$; (c) $X_n / Y_n \xrightarrow{d} Z/c$ if $c \neq 0$.

Proof. (a) For a continuity point z of $Z + c$ and small ϵ such that $z \pm \epsilon$ are continuity points,

$$P(X_n + Y_n \leq z) \leq P(X_n \leq z - c + \epsilon) + P(|Y_n - c| \geq \epsilon),$$

and similarly $P(X_n + Y_n \leq z) \geq P(X_n \leq z - c - \epsilon) - P(|Y_n - c| \geq \epsilon)$. Let $n \to \infty$, then $\epsilon \downarrow 0$. (b) First $c = 0$: $P(|Y_n X_n| \geq \epsilon) \leq P(|X_n| > k) + P(|Y_n| \geq \epsilon/k)$; the first term is small for large

k uniformly in large n (tightness of X_n), the second $\to 0$. In general $Y_nX_n = (Y_n - c)X_n + cX_n$; apply the case $c = 0$ and (a). (c) $1/Y_n \xrightarrow{p} 1/c$ by Theorem 5.5; apply (b). ■

Example 5.2 (Studentized mean)

For any population with $0 < \sigma^2 < \infty$, $\frac{\sqrt{n}(\bar{X}_n-\mu)}{S_n} = \frac{\sqrt{n}(\bar{X}_n-\mu)/\sigma}{S_n/\sigma} \xrightarrow{d} N(0,1)$. Hence $\bar{X}_n \pm z_{\alpha/2}S_n/\sqrt{n}$ is an asymptotic $1-\alpha$ interval for μ, without normality. Under normality $t(n-1) \to N(0,1)$ is a special case.

Theorem 5.7 (Delta method)

If $\sqrt{n}(X_n-\theta) \xrightarrow{d} N(0,\sigma^2)$ and g is differentiable at θ, then $\sqrt{n}(g(X_n)-g(\theta)) \xrightarrow{d} N(0, g'(\theta)^2\sigma^2)$. In k dimensions, if $\sqrt{n}(\mathbf{X}_n - \theta) \xrightarrow{d} N(\mathbf{0}, \Sigma)$, then $\sqrt{n}(g(\mathbf{X}_n) - g(\theta)) \xrightarrow{d} N(0, \nabla g(\theta)'\Sigma\nabla g(\theta))$.

Proof. By differentiability, $g(x) - g(\theta) = (g'(\theta) + r(x))(x - \theta)$ with $r(x) \to 0$ as $x \to \theta$ ($r(\theta) := 0$). Since $X_n - \theta = \frac{1}{\sqrt{n}} \cdot \sqrt{n}(X_n - \theta) \xrightarrow{p} 0$ by Slutsky, $r(X_n) \xrightarrow{p} 0$. Then $\sqrt{n}(g(X_n) - g(\theta)) = (g'(\theta) + r(X_n))\sqrt{n}(X_n - \theta) \xrightarrow{d} g'(\theta)N(0,\sigma^2)$ by Slutsky. The vector case is identical with a gradient. ■

Example 5.3

(a) For Bernoulli data, $\sqrt{n}(\hat{p}-p) \xrightarrow{d} N(0, p(1-p))$, so with $g(p) = \arcsin\sqrt{p}$, $g'(p) = \frac{1}{2\sqrt{p(1-p)}}$ gives $\sqrt{n}(\arcsin\sqrt{\hat{p}} - \arcsin\sqrt{p}) \xrightarrow{d} N(0, \frac{1}{4})$: a variance-stabilizing transformation. (b) For Poisson data, $2\sqrt{\bar{X}}$ has asymptotic variance $1/n$ whatever λ. (c) For the log-odds $\log\frac{\hat{p}}{1-\hat{p}}$, the asymptotic variance is $\frac{1}{np(1-p)}$.

Theorem 5.8 (Sample quantiles)

Let F have density f continuous and positive at $\xi_\alpha = F^{-1}(\alpha)$, and let $r_n/n \to \alpha$ with $\sqrt{n}(r_n/n - \alpha) \to 0$. Then $\sqrt{n}(X_{(r_n)} - \xi_\alpha) \xrightarrow{d} N(0, \frac{\alpha(1-\alpha)}{f(\xi_\alpha)^2})$.

Proof. $P(X_{(r_n)} \le \xi_\alpha + t/\sqrt{n}) = P(\#\{X_i \le \xi_\alpha + t/\sqrt{n}\} \ge r_n)$. The count is $B(n, p_n)$ with $p_n = F(\xi_\alpha + t/\sqrt{n}) = \alpha + f(\xi_\alpha)t/\sqrt{n} + o(n^{-1/2})$. By the CLT for binomials with $p_n \to \alpha$, the probability tends to $P(Z \ge \frac{-f(\xi_\alpha)t}{\sqrt{\alpha(1-\alpha)}}) = \Phi(\frac{f(\xi_\alpha)t}{\sqrt{\alpha(1-\alpha)}})$. ■

For the median of $N(\mu,\sigma^2)$, $\alpha = \frac{1}{2}$ and $f(\mu) = 1/(\sigma\sqrt{2\pi})$, so the asymptotic variance is $\pi\sigma^2/(2n)$, larger than σ^2/n for $\bar{X}$.

5.4 Approximation by simulation

When neither exact nor asymptotic theory is convenient, the distribution of a statistic T can be approximated by **Monte Carlo**: generate B independent samples from the model, compute $T_1, \ldots, T_B$, and use their empirical distribution.

- **Generating random numbers.** From $U \sim U(0,1)$, $F^{-1}(U) \sim F$ (Section 4.1). For example $-\beta\log(1-U) \sim \text{Exp}(\beta)$, $\log\frac{U}{1-U}$ is standard logistic, $\mu + \sigma\Phi^{-1}(U) \sim N(\mu,\sigma^2)$. Sums, ratios and so on of generated variables give gamma, χ^2, t, F variates.
- **Accuracy.** An estimated probability $\hat{p} = \frac{1}{B}\sum I(T_b \in A)$ is binomial, with standard error $\sqrt{p(1-p)/B}$. An estimated mean $\bar{T}$ has standard error $s_T/\sqrt{B}$. By the CLT, $\hat{p} \pm$

$2\sqrt{\hat{p}(1-\hat{p})/B}$ is an approximate 95% band for the simulation error; $B = 10^4$ gives about ± 0.004 for $p \approx 0.05$.

- **Uses.** Checking the accuracy of asymptotic approximations at a given n (e.g. coverage of $\bar{X} \pm z_{\alpha/2} S/\sqrt{n}$ for skewed populations), and approximating quantiles of statistics with intractable distributions.